

GERALDTON, ON, Canada

WMO: 718340

Lat: 49.783N

Lon: 86.931W

Elev: 1143

StdP: 14.10

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.1	(c) -25.4	(d) -39.3	(e) 0.6	(f) -31.0	(g) -33.4	(h) 0.9	(i) -25.1	(j) 19.9	(k) 12.8	(l) 18.5	(m) 11.0	(n) 2.2	(o) 250	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 82.6	(d) 66.8	(e) 79.6	(f) 64.6	(g) 76.8	(h) 63.1	(i) 69.5	(j) 78.7	(k) 67.3	(l) 75.6	(m) 65.4	(n) 73.0	(o) 11.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	101.2	74.1	64.2	93.8	71.4	62.4	88.0	69.3	34.3	78.4	32.5	75.7	30.9	73.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
19.5	17.9	15.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-40.4	88.3	6.6	3.4	-45.1	90.7	-49.0	92.7	-52.6	94.6	-57.4	97.1		
			-40.4	73.3	6.3	1.9	-44.9	74.7	-48.6	75.8	-52.1	76.8	-56.7	78.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	0.1	4.0	17.2	32.3	47.2	58.9	63.7	61.4	52.6	39.2	23.8	8.8
	DBStd	24.56	14.58	13.58	13.94	10.37	8.89	7.37	5.99	6.19	8.32	8.61	11.98	14.11
	HDD50	7052	1546	1288	1018	534	164	14	1	2	67	355	787	1276
	HDD65	11349	2011	1708	1482	981	556	207	98	148	379	801	1237	1741
	CDD50	1304	0	0	1	3	76	280	424	355	145	20	0	0
	CDD65	125	0	0	0	0	4	23	56	35	7	0	0	0
	CDH74	1275	0	0	0	1	68	312	522	314	57	1	0	0
	CDH80	236	0	0	0	0	11	64	104	51	6	0	0	0
(14) Wind	WSAvg	7.0	6.5	6.8	7.2	7.5	7.7	7.0	6.9	6.7	7.0	7.5	7.3	6.5
(15) Precipitation	PrecAvg	29.8	1.7	1.2	1.6	2.0	2.9	3.6	3.3	3.2	3.4	2.8	2.2	1.7
	PrecMax	41.6	4.8	2.1	3.1	4.4	5.7	8.0	7.0	6.4	7.0	5.1	4.8	2.9
	PrecMin	22.7	0.7	0.6	0.9	0.7	1.1	1.1	1.1	1.1	0.5	0.7	1.1	0.9
	PrecStd	4.5	0.8	0.4	0.5	0.7	1.2	1.5	1.6	1.5	1.7	1.3	0.8	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	41.2	56.1	67.1	80.5	86.3	87.1	85.0	79.9	69.1	54.0	38.5
		MCWB	33.0	37.7	48.0	50.8	60.6	67.7	70.0	69.2	65.6	57.6	48.0	36.1
	2%	DB	32.3	35.0	47.3	59.7	74.9	80.9	82.8	80.8	74.7	62.6	47.7	35.4
		MCWB	31.1	32.4	40.6	45.8	58.1	64.1	67.5	66.5	63.4	55.2	44.0	33.4
	5%	DB	29.0	30.7	41.5	54.6	70.0	77.4	79.5	77.5	70.2	57.6	43.1	32.6
		MCWB	27.6	28.5	35.7	42.8	55.1	61.9	65.1	64.8	61.0	52.4	40.0	31.3
	10%	DB	23.8	26.4	36.8	49.4	64.6	73.8	76.6	74.0	66.3	53.2	39.4	29.6
		MCWB	22.4	24.5	32.8	39.8	52.2	60.1	63.5	62.9	59.5	48.9	37.4	28.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	37.2	49.5	52.4	65.3	70.9	72.8	71.8	68.3	60.6	49.7	36.8
		MCDB	34.4	40.0	54.7	63.6	75.1	83.8	83.6	80.7	76.1	66.4	52.4	38.0
	2%	WB	31.3	32.6	41.1	47.0	60.2	67.0	69.9	68.9	65.3	56.3	44.4	33.7
		MCDB	32.4	34.7	46.1	57.9	70.3	76.9	79.0	77.4	71.4	61.3	47.2	34.9
	5%	WB	27.6	28.8	36.6	43.8	56.8	64.2	67.4	66.8	62.9	52.7	40.5	31.4
		MCDB	29.0	30.6	40.3	52.4	67.1	71.9	75.9	74.3	68.1	57.0	42.5	32.6
	10%	WB	22.5	24.5	33.0	40.6	53.6	62.1	65.3	64.7	60.2	49.0	37.3	28.4
		MCDB	23.8	26.4	36.7	49.2	62.8	70.4	73.2	71.9	65.5	52.8	39.4	29.6
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	23.5	24.1	22.2	22.5	21.7	20.6	20.0	18.5	13.8	13.6	16.4
		MCDBR	17.5	21.6	24.0	30.9	32.0	28.1	24.9	24.8	24.1	21.8	16.8	15.3
	5% WB	MCWBR	16.5	19.0	17.6	19.0	17.1	14.2	12.6	13.1	14.4	14.6	13.2	14.3
		MCDWR	16.6	19.0	20.4	27.4	25.6	22.3	21.4	21.1	19.5	19.7	15.1	15.3
(40) Clear-Sky Solar Irradiance	taub		0.249	0.259	0.296	0.346	0.351	0.366	0.389	0.366	0.347	0.306	0.280	0.272
		taud	2.334	2.315	2.305	2.335	2.426	2.411	2.371	2.437	2.490	2.574	2.497	2.271
	Ebn at Noon		259	285	290	285	288	283	274	276	271	266	247	226
		Edh at Noon	21	27	33	36	35	36	36	33	28	22	18	19
(44) All-Sky Solar Radiation	RadAvg		373	729	1159	1538	1650	1811	1783	1490	1055	618	318	259
	RadStd		45	55	84	125	107	138	128	117	84	89	35	37

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page